

Isabella Li

UW Seattle | Isabella.y.li@outlook.com | github.com/lzzyL234 | Computer Engineering, Class of 2029

Computer Engineering student at UW's Allen School with hands-on experience in machine learning, computer vision, embedded systems, and data-center hardware. Interned at Asia Vital Components on cooling and thermal systems for a hyperscale cloud provider's AI accelerator programs. Built AnemoDx, a computer-vision hemoglobin screening system that won the WA-01 Congressional App Challenge and co-authored a clinical machine-learning paper on ICU outcomes for 8,070 trauma patients.

EDUCATION

University of Washington, Seattle

Expected 2029

B.S. in Computer Engineering (Direct Admit), Paul G. Allen School of Computer Science & Engineering

SKILLS

Languages: Python, C++, Java, SQL, CircuitPython

ML/Data: PyTorch, TensorFlow, scikit-learn, CNNs, clustering, time-series analysis, data visualization, Qiskit

Hardware/Embedded: Raspberry Pi, ESP32-CAM, PCB design, KiCad, serial communication protocols, Fusion 360, 3D printing

Tools: Git/GitHub, Excel, Figma

EXPERIENCE

ASIA VITAL COMPONENTS

Redmond, WA | Jul – Sep 2026

Leading global thermal management & data center hardware solutions provider | TWSE: 3017

Intern, AI Infrastructure Programs

- Supported AI accelerator infrastructure programs for a hyperscale cloud provider across liquid cooling, air cooling, and thermal systems
- Coordinated weekly with cloud-provider, Taiwan-based, and partner engineering teams on technical requirements, design changes, and manufacturing readiness
- Built a parts calculator covering 156 component line items across 15 project groupings to calculate rack- and tray-level quantities and costs
- Developed manufacturing lookup tools covering 88 manufacturing process records and 26 equipment entries to support cycle-time, staffing, and equipment-cost analysis
- Built a timeline tool tracking 9 milestones from tooling trials to mass production; analyzed component sourcing, capacity, lead times, and production schedules to assess supply-chain risks

UW SENSORS, ENERGY, AND AUTOMATION LAB

Seattle, WA | Sep 2026 – Present

Student Researcher

Selected for undergraduate research in sensing and hardware-software systems

ART OF PROBLEM SOLVING

Redmond, WA | Jul 2024 – Present

Teaching Assistant

Teach math and language arts; coach students through problem solving and written feedback.

PROJECTS

AnemoDx: Computer-Vision Hemoglobin Screening | Demo: [AnemoDx Video](#) Jul 2025

- Won the WA-01 Congressional App Challenge for developing a low-cost, non-invasive hemoglobin screening system using fingernail images
- Developed and evaluated a python computer-vision/ML pipeline on a clinical dataset of ~250 patients, achieving 1.36 g/dL mean absolute error in hemoglobin estimation
- Built the end-to-end scanning device, integrating a 3D-printed enclosure, camera, controlled LED illumination, and embedded electronics for standardized image capture.

Hack Club Hardware Hackathon | GitHub: [HackCam](#) Shenzhen, China | Jul 2026

- Selected for a fully funded hardware hackathon after completing 60+ hours of independent hardware projects and technical preparation
- Designed and built HackCam, a portable Raspberry Pi 5 camera integrating a camera module, SPI display, rechargeable LiPo battery, and physical controls, in a 3-day team sprint
- Developed Python firmware using Picamera2, Pygame, NumPy, spidev, and gpiozero to stream a live camera preview to the SPI display, process GPIO shutter-button input, capture and save photos, and render UI

RESEARCH

Daily Sequential Organ Failure Assessment (SOFA) Score Time Series Analysis Towards Early Outcome Prediction in ICU Trauma Patients Sep 2026

So-author | Submitted to IEEE BigData 2026

- Built a Python pipeline using K-means clustering to analyze 14-day SOFA trajectories for 8,070 ICU trauma patients
- Identified 15 distinct patient trajectory groups, with 14-day mortality ranging from 0.0% to 81.4% across clusters
- Analyzed Day-28 mortality and discharge timing among long-stay survivors and generated manuscript figures for publication with UW Tacoma and UW Medicine researchers

Creativity, Subtext, and Shared Context in AI Communication Jul 2026

- Built a Python/Gemini API experiment testing how language models generate and interpret indirect clues based on shared context; analyzed instruction noncompliance and reuse of explicit descriptive language.

A Comparative Analysis of Quantum and Classical Support Vector Machines on Complex Data for Customer Churn Prediction Jul 2025

- Benchmarked QSVM, SVM, and Random Forest models for customer churn classification using Qiskit and scikit-learn, applying feature selection, preprocessing, quantum kernels, and repeated evaluation; presented findings at the UCSB Summer Research Academies symposium.

LEADERSHIP & AWARDS

The Harvard Undergraduate Research Journal Research Competition Nov 2025 US and North America Champion; Global Finalist; Semifinalist (Original Research and Research Proposal).

Founder, Society of Women Engineers SWENext Club, Redmond High 2023 – 2026 Established the first SWENext chapter in the Lake Washington School District, secured \$1,000+ in funding and led mentorship, speaker, outreach, and nonprofit partnerships reaching 2,000+ K–12 students.

Co-Host, Youth Talk, Chinese Radio Seattle 2023 – 2026 Shared student experiences, taught situational English, and interviewed community leaders.